CPS 843 (CP 8307) Problem Set 4

Due Monday, November 17th, @ 11:59pm
(5 assignments in total, final grade is evaluated based on the best 4 out of 5)

Purpose

- Familiar with the algorithm in computer vision
- Understand the basic concepts of imaging process and single-view geometry

Requirements

Part 1:

Problem 1. (1) Write the general form of a finite projective camera's intrinsic parameter matrix **K** and give the meaning of each parameter in K. (2) If we assume the pixels are rectangular, how many degrees of freedom does the projection matrix have, and why? (3) Given the projection matrix $\mathrm{P} = [\mathrm{M} \mid \mathbf{p}_4]$, show how to recover the internal and external parameters from the projection matrix. (4) Verify that the image of a point at infinity is only affected by the sub-matrix **M**. (8 points)

Problem 2. (1) Given the projection matrix **P**, show how to compute the optical center of the camera. (2) Verify that the first three columns of the projection matrix correspond to the vanishing points of the X, Y, and Z axes of the world system. (3) Verify that the last row of the projection matrix corresponds to the principal plane of the camera. (4) What is the geometric meaning of the last row of **P** and why? (8 points)

Part 2: (7 points)

Data:
Take a set of images (about 10-15 frames) of a scene with more than 60% overlap between any two adjacent frames. To achieve better results, the images should be taken by rotating the camera around its optical center.

Software:
You may choose any package below:
1. AutoStitch: http://matthewalunbrown.com/autostitch/autostitch.html
2. OpenPano: https://github.com/ppwwyyxx/OpenPano
3. Hugin: https://hugin.sourceforge.io/

Work to Do

1. Read the paper "Automatic Panoramic Image Stitching using Invariant Features," by M. Brown and D. Lowe, which can be downloaded from the above link.
2. Read the paper for SIFT feature matching "Distinctive image features from scale-invariant keypoints," which can be downloaded from http://www.cs.ubc.ca/~lowe/keypoints/

3. You may also read the tutorial of "OpenPano" https://ppwwyyxx.com/blog/2016/How-to-Write-a-Panorama-Stitcher/
4. Download the source code or executable file via one of the above links. Use the software to stitch the images you took to generate a panoramic view.

Report:

- A brief technical overview of SIFT feature matching and image stitching in your own words based on your understanding (within 1 page).
- Show 6 sample original images and the stitching result you obtained.
- Make a brief analysis and discussion of your result based on your understanding.

Part 3: (2 points)

- A brief progress report of your final project in one short paragraph.

Requirements

- The assignment is due on Monday, November 17th @ 11:59pm. You have 24 hours of grace time without penalty. Submissions after the grace period will not be accepted.
- Submit all your work in a single PDF file through D2L (multiple submissions are allowed, but only the last submission will be considered for evaluation).
- Please resize all images properly in line with the text of your report.
- Submit the source code in line with the report of each part.
- You can directly use any available functions of Matlab for your work. Other programming languages and packages are also allowed.
- Complete the report by yourself. We will use Turnitin® for similarity check.

TA for HW#4:

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You may contact the TA directly if you have any questions about this assignment.